

Algebra I Functions and Modeling Cheatsheet

Module 10

Function Rule

A function assigns each input exactly one output.

$$x \rightarrow f(x)$$

Vertical line test:

a graph is a function if every vertical line hits it at most once

Function Notation

$$f(x) = 2x - 7$$

$$f(5) = 2(5) - 7 = 3$$

The input is the value substituted for x .

Domain and Range

domain = set of input values

range = set of output values

For tables, list the x -values for the domain and the y -values for the range.

Representations

Functions can be shown with:

equations, tables, graphs, verbal descriptions

Connections:

table differences $\leftrightarrow$ rate of change

graph intercepts $\leftrightarrow$ starting values or zeros

Arithmetic Sequences

$$a_n = a_1 + (n - 1)d$$

Constant difference:

$$d = a_n - a_{n-1}$$

Arithmetic sequences are linear functions of the term number.

Simple Exponential Growth

$$y = a(1 + r)^t$$

Growth:

$$r > 0$$

Decay:

$$y = a(1 - r)^t, \quad 0 < r < 1$$

Exponential tables have a constant ratio.

Model Choice

Pattern	Table	Model
constant difference	add same amount	linear
constant ratio	multiply by same amount	exponential
symmetry with turning point	second differences constant	quadratic

Quick Checks

- Inputs cannot repeat with different outputs in a function.
- Linear models add by a constant amount.
- Exponential models multiply by a constant amount.
- Interpret variables with units when modeling.